

A Dimension-Four Twisted-Convolution Identity from a Shintani Ray Unit

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Abstract

We study the rank condition that turns canonical Shintani–Faddeev real-multiplication values into ghost projectors for Weyl–Heisenberg symmetric informationally complete measurements. We give exact Zak-rank, exterior-square, and parity-moment formulations for the canonical family. In dimension four, all 36 minors of one explicitly defined normalized ghost matrix are divisible by the common factor $x^2 - \sqrt{3 + \sqrt{5}}x + 1$. Using the Shintani–Kopp Kronecker-limit theorem, we identify the modulus, labeled embeddings, identity ray class, quarter characteristic, positive stabilizer, multiplier, and partial-zeta sign. The analytic class-number formula for $\mathbb{Q}(\sqrt{5}, \sqrt{\phi})$ then gives $x^2 = \phi + \sqrt{\phi}$. Consequently the ghost has rank one. The identity-twist equation is the $\lambda = 1$ TCC shift, and complex conjugation proves the paired $\lambda = 0$ shift. Thus the formal dimension-four Twisted Convolution Conjecture holds unconditionally.

1 Canonical setup

For $d \geq 4$, put

$$Q_d = \langle 1, 1 - d, 1 \rangle, \quad L_d = \begin{pmatrix} d-1 & -1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$\text{disc}(Q_d) = (d+1)(d-3), \quad L_d^2 + L_d + I = dL_d,$$

so $L_d^3 \equiv I \pmod{d}$. Its positive quadratic fixed point is

$$\beta_d = \frac{d-1 + \sqrt{(d+1)(d-3)}}{2}.$$

For $D_{p,q} = \tau^{pq} X^p Z^q$, $\tau = -e^{\pi i/d}$, write

$$P = \frac{1}{d} \sum_{\mathbf{p}} a_{\mathbf{p}} D_{\mathbf{p}}.$$

The equation $P^2 = P$ is a finite twisted-convolution equation for the coefficients. In even dimensions, the representative-wrap signs of the displacement operators are retained. The principal-ghost and formal TCC constructions motivating this normalization are those of Appleby–Flammia–Kopp [2]; software is not used as their mathematical source.

2 Finite rank certificates

After a scalar shift, shear, and partial Fourier transform, canonical twisted convolution is equivalent to rank one of a $d \times d$ matrix K_d . Thus it is equivalent to the vanishing of all 2-minors.

For any complex matrix K ,

$$\Delta_2(K) = \frac{1}{2} \left((\text{Tr } K^\dagger K)^2 - \text{Tr}((K^\dagger K)^2) \right) = \sum_{\substack{|I|=2 \\ |J|=2}} |\det K_{I,J}|^2.$$

Hence $\Delta_2(K) = 0$ if and only if $\text{rank } K \leq 1$.

The ghost also obeys parity-Hermiticity $G^\dagger = PGP$. Therefore $J = PG$ is Hermitian and the rank-one certificate becomes

$$\text{Tr } J^4 = (\text{Tr } J^2)^2.$$

The fourth-moment identity is used only as an equivalent diagnostic here; the dimension-four proof below uses the complete exact minor certificate.

3 Dimension four and conventions

Let

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad \beta = \phi^2, \quad \tau = -e^{\pi i/4}.$$

To prevent convention ambiguity, define the function used here directly by its gamma ratio:

$$S_2^{\text{here}}(z \mid \omega_1, \omega_2) := \frac{\Gamma_2(\omega_1 + \omega_2 - z \mid \omega_1, \omega_2)}{\Gamma_2(z \mid \omega_1, \omega_2)}.$$

This definition is decisive. Using Kopp's $\text{Sin}_{2,\text{Kopp}}$ instead in the display defining x changes x^2 from u to $4/u$, because the prefactor $\sqrt{2}$ is not inverted.

The normalized signed double-sine table is

$$\begin{pmatrix} \sqrt{5} & -x & 1 & -x^{-1} \\ -x^{-1} & -x^{-1} & -x^{-1} & -x \\ 1 & -x^{-1} & 1 & x \\ -x & -x^{-1} & x & -x \end{pmatrix}.$$

The zero characteristic is exceptional and is replaced by $\sqrt{5}$ before applying the displacement chirp. Every table entry is then divided by $\sqrt{5}$ in forming the normalized Weyl overlaps; in particular the exceptional entry gives $a_0 = 1$.

Define the matrix used in this paper, without reference to software, by

$$K_{r,c}(x) = \frac{1}{4\sqrt{5}} \sum_{\substack{0 \leq a, b < 4 \\ r \equiv c+a \pmod{4}}} T_{a,b}(x) \tau^{ab} i^{bc}, \quad 0 \leq r, c < 4,$$

where $T(x)$ is the displayed table and $\tau = -(1+i)/\sqrt{2}$. Reduction of $r = c + a$ modulo four is performed before evaluating the entry; the factor τ^{ab} retains the even-dimensional displacement sign convention. This formula is the complete definition of the matrix whose minors are certified below.

Lemma 1 (Weyl reconstruction). *Let $\nu_{a,b} = T_{a,b}(x)$ for the sixteen entries of the displayed table, and put $a_{a,b} = \nu_{a,b}/\sqrt{5}$. Thus $a_{0,0} = 1$. With*

$$D_{a,b} e_c = \tau^{ab} i^{bc} e_{c+a} \quad (a, b, c \in \mathbb{Z}/4\mathbb{Z}),$$

the operator reconstructed from these sixteen normalized ghost overlaps is

$$\Pi_I = \frac{1}{4} \sum_{a,b \bmod 4} a_{a,b} D_{a,b}, \quad (\Pi_I)_{r,c} = \frac{1}{4\sqrt{5}} \sum_{r \equiv c+a} T_{a,b} \tau^{ab} i^{bc} = K_{r,c}(x).$$

Moreover,

$$\mathrm{Tr} \Pi_I = \frac{T_{0,0}}{\sqrt{5}} = 1.$$

Proof. The displayed action of $D_{a,b}$ gives the matrix-entry formula term by term. On taking the trace, the sum over c annihilates every $b \neq 0$, while $a = 0$ is forced on the diagonal. Hence only $T_{0,0} = \sqrt{5}$ remains. $\square$

Proposition 2. *Every one of the 36 two-by-two minors of the dimension-four matrix $K(x)$ lies in the principal ideal*

$$\left(x^2 - \sqrt{3 + \sqrt{5}}x + 1\right)$$

of $\mathbb{Q}(i, \sqrt{2}, \sqrt{5})[x, x^{-1}]$.

Proof. The exact Laurent expansion has 34 formally nonzero minors and two structurally zero minors. Reduce each power using

$$x^{-1} = t - x, \quad x^2 = tx - 1, \quad t = \sqrt{3 + \sqrt{5}} = \frac{\sqrt{2} + \sqrt{10}}{2}.$$

Every remainder $A + Bx$ is zero. More explicitly, for every row pair I and column pair J , the supplementary certificate records a Laurent polynomial $M_{I,J}$ and a polynomial $Q_{I,J}$ satisfying

$$x^2 M_{I,J}(x) = (x^2 - tx + 1)Q_{I,J}(x)$$

coefficientwise in the genuine $\mathbb{Q}$ -basis

$$1, \sqrt{2}, \sqrt{5}, \sqrt{10}$$

and separately in real and imaginary parts. Thus the certificate is verified by rational arithmetic only. It contains all 36 minors, including the two zero quotients, rather than a Boolean summary. $\square$

It remains to prove $x + x^{-1} = \sqrt{3 + \sqrt{5}}$. Shift and reflection give

$$x = \sqrt{2} \frac{S_2^{\text{here}}(\beta/4 \mid \beta, 1) S_2^{\text{here}}(1/4 \mid \beta, 1)}{S_2^{\text{here}}((\beta + 1)/4 \mid \beta, 1)}.$$

4 The modulus-four ray field

Let $K = \mathbb{Q}(\sqrt{5})$ and $\mathcal{O}_K = \mathbb{Z}[\phi]$. Since 2 is inert, there are $|(\mathcal{O}_K/4)^\times| = 12$ finite residue units. The images of $\{\pm\phi^n\}$, together with the sign at ∞_2 , have order 12. Since $h_K = 1$, the ray exact sequence therefore gives

$$|\mathrm{Cl}_{(4)\infty_2}(K)| = 2.$$

Label the embeddings by $\infty_1(\sqrt{5}) = +\sqrt{5}$ and $\infty_2(\sqrt{5}) = -\sqrt{5}$. Set $L = K(\sqrt{\phi})$. Its relative discriminant is (4), and the second real place ramifies because $\phi' < 0$. Hence L is the full ray field of modulus $(4)\infty_2$: the quadratic conductor–discriminant theorem gives its conductor, and the previously computed ray group has order two.

Put $u = \phi + \sqrt{\phi}$, and define the nontrivial $\sigma \in \mathrm{Gal}(L/K)$ by $\sigma(\sqrt{\phi}) = -\sqrt{\phi}$. Then $u^\sigma = \phi - \sqrt{\phi} = u^{-1}$, so

$$N_{L/K}(u) = 1, \quad \mathrm{Tr}_{L/K}(u) = 1 + \sqrt{5}.$$

5 Class number, units, and regulator

The field L has signature $(2, 1)$ and discriminant -400 . Its Minkowski bound is $15/(2\pi) < 2.4$. There is no ideal of norm two: after establishing $\mathcal{O}_L = \mathbb{Z}[s]$, reduction of its defining polynomial gives

$$X^4 - X^2 - 1 \equiv (X^2 + X + 1)^2 \pmod{2},$$

so every prime above two has residue degree two and norm four. Hence $h_L = 1$.

Write $s = \sqrt{\phi}$, so $s^4 - s^2 - 1 = 0$. The power-basis discriminant is -400 , equal to the certified field discriminant below; hence $\mathcal{O}_L = \mathbb{Z}[s]$. An exact Buchmann class-group/unit computation in PARI/GP 2.15.4 [6] returns fundamental units $s^3 - s$ and $-s^2 + s$. Since

$$s^3 - s = s^{-1}, \quad -s^2 + s = -u^{-1},$$

these generate the same rank-two lattice as s and $u = s^2 + s$; the exponent-change matrix has determinant one. Consequently

$$\mathcal{O}_L^\times = \{\pm s^m u^n : m, n \in \mathbb{Z}\}, \quad R_L = \log \phi \log u.$$

The exact computation also returns class number one, ray structures $[2]$ and $[2, 2]$, and the same regulator. The output `bnfcertify(B)=1` certifies the class group, regulator, and fundamental-unit computation unconditionally. The complete input and output are supplied in `scripts/referee_pari_audit.gp` and `certificates/pari-audit.txt`.

For the nontrivial quadratic character χ of L/K ,

$$L(s, \chi) = \frac{\zeta_L(s)}{\zeta_K(s)}.$$

For a number field F , the analytic class-number formula at zero is

$$\zeta_F(s) = -\frac{h_F R_F}{w_F} s^{r_1+r_2-1} + O(s^{r_1+r_2}).$$

Since $h_K = h_L = 1$, $w_K = w_L = 2$, $R_K = \log \phi$, and the respective orders of vanishing are one and two, division of these leading terms yields

$$L'(0, \chi) = \log u.$$

6 The Kronecker-limit bridge

Put

$$B = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}^3 = \begin{pmatrix} 21 & -8 \\ 8 & -3 \end{pmatrix}, \quad \mathbf{r} = \begin{pmatrix} 0 \\ 1/4 \end{pmatrix}.$$

Take Kopp's modulus $\mathfrak{m} = 4\mathcal{O}_K$, the identity ray class $\mathfrak{A}$, $\mathfrak{b} = \mathcal{O}_K$, and $\alpha = 4$. Then

$$\mathfrak{b}\mathfrak{m} = 4(\beta\mathbb{Z} + \mathbb{Z}), \quad \alpha(r_2\beta - r_1)\mathcal{O}_K = \beta\mathcal{O}_K = \mathcal{O}_K \in \mathfrak{A},$$

and $r_2\beta' - r_1 = \beta'/4 > 0$. Thus these choices satisfy every ideal and positivity hypothesis in Kopp's Theorem 1.1.

The matrix B fixes β , belongs to $\Gamma_{\mathbf{r}}$ because $B\mathbf{r} - \mathbf{r} = (-2, -1)^\top$, and is the positive generator of that stabilizer; its expanding eigenvalue is β^3 . The fiber above $\mathfrak{A}$ under

$$\mathrm{Cl}_{(4)\infty_1\infty_2}(K) \longrightarrow \mathrm{Cl}_{(4)\infty_2}(K)$$

has order two, so Kopp's exponent is $n = 2/2 = 1$.

Kopp's sign class $\mathfrak{R}$ is the nonidentity element of the order-two ray group. Here is a direct ray calculation of that fact. The kernel created by adding ∞_1 would be trivial only if there were a unit congruent to 1 modulo 4, positive at ∞_2 , and negative at ∞_1 . Every unit is $\pm\phi^n$. For $0 \leq n < 6$, the residues of $\phi^n = a_n + b_n\phi$ modulo 4 are

$$(a_n, b_n) = (1, 0), (0, 1), (1, 1), (1, 2), (2, 3), (3, 1),$$

and $\phi^6 \equiv 1 \pmod{4}$. The only positive-sign unit in the residue class 1 is therefore positive at both real places; multiplying by -1 changes the residue class. Thus the kernel has order two, and its nonidentity sign class is $\mathfrak{R}$. Equivalently, its Artin image is the inertia generator at the newly ramified real place. Hence, for its nontrivial quadratic character χ ,

$$Z_{(4)\infty_2}(s, \mathfrak{A}) = \zeta(s, \mathfrak{A}) - \zeta(s, \mathfrak{R}\mathfrak{A}) = L(s, \chi).$$

For $\mathbf{p} = (0, 1)$, the Rademacher invariant of B is zero: $s(21, 8) = -1/16$ gives $18/8 - 3 - 12(-1/16) = 0$. The principal-ghost phase is

$$\Phi_{\mathbf{p}} = \xi_4^{-1} = e^{3\pi i/4}, \quad \Phi_{\mathbf{p}}^2 = -i = (\psi^{-2}\chi_{\mathbf{r}}^{-1})(B).$$

Explicitly, the zero Rademacher invariant gives $\psi^2(B) = 1$. Substitution into Kopp's defining theta-character formula (before any even-denominator simplification) gives

$$\chi_{\mathbf{r}}(B) = \exp\left(2\pi i \frac{1}{2} \left(-\frac{28}{4} + \frac{152}{16}\right)\right) = \exp(2\pi i 5/4) = i.$$

Thus $\chi_{\mathbf{r}}^{-1}(B) = -i$. This is also exactly the specialization of the phase relation of Appleby–Flammia–Kopp.

Lemma 3 (Convention-matched cocycle value). *For $\mathbf{p} = (0, 1)$, AFK's phase and Kopp's modular cocycle give*

$$\nu_{\mathbf{p}} = \Phi_{\mathbf{p}} \operatorname{shin}_B^{\mathbf{r}}(\beta) = -x,$$

where, before shift/reflection simplification,

$$x = \prod_{z \in \{1+\beta/4, 1/4, 1+(3\beta-1)/4\}} S_2^{\text{here}}(z \mid \beta, 1).$$

Proof. Write $C = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}$, so $B = C^3$. Modulo four the characteristic orbit under the three cocycle factors is

$$(0, 1) \mapsto (3, 0) \mapsto (1, 3) \mapsto (0, 1).$$

Insert these three characteristics in the cocycle law for C^3 and then in AFK Definition 1.32. Kopp's S -generator formula uses $\operatorname{Sin}_{2, \text{Kopp}} = \Gamma_2(z)/\Gamma_2(\beta + 1 - z)$, whereas our definition is its reciprocal. The three factors are consequently, in orbit order,

$$S_2^{\text{here}}(1 + \beta/4), \quad S_2^{\text{here}}(1/4), \quad S_2^{\text{here}}(1 + (3\beta - 1)/4),$$

all with periods $(\beta, 1)$. The integral translations in the three characteristics contribute exponent $1 + 1 + 3 = 5$; after multiplication by $\Phi_{(0,1)} = \xi_4^{-1}$ their phase is $(-1)^5 = -1$. This gives the asserted signed value. This calculation is precisely the specialization of AFK Definition 1.32 and its cocycle factorization, with the gamma-ratio convention written out rather than left implicit. $\square$

Because S_2^{here} is the reciprocal of the Shintani–Kopp double sine, its shift and reflection identities give

$$x = \sqrt{2} \frac{S_2^{\text{here}}(\beta/4 \mid \beta, 1) S_2^{\text{here}}(1/4 \mid \beta, 1)}{S_2^{\text{here}}((\beta + 1)/4 \mid \beta, 1)}.$$

Squaring Lemma 3, and using the multiplier computed above, gives line by line

$$x^2 = \nu_{(0,1)}^2 = \Phi_{(0,1)}^2 \operatorname{shin}_B^{\mathbf{r}}(\beta)^2 = (\psi^{-2} \chi_{\mathbf{r}}^{-1})(B) \operatorname{shin}_B^{\mathbf{r}}(\beta)^2 = \exp Z'_{(4)\infty_2}(0, \mathfrak{A}) = \exp L'(0, \chi) = u. \quad (\text{KL})$$

Theorem 4 (Dimension-four rank theorem). *The explicitly defined matrix $K(x)$ has rank exactly one.*

Proof. By (KL), $x^2 = u$. Since $u + u^{-1} = 1 + \sqrt{5}$,

$$(x + x^{-1})^2 = 3 + \sqrt{5}.$$

The defining reciprocal double-sine value is positive, hence $x + x^{-1} = \sqrt{3 + \sqrt{5}}$. The common minor factor vanishes, so all 36 minors vanish by Proposition 2. Thus $\operatorname{rank} K(x) \leq 1$. Moreover, the explicit definition gives

$$K_{0,0}(x) = \frac{\sqrt{5} + 1 - x - x^{-1}}{4\sqrt{5}} = \frac{\sqrt{5} + 1 - \sqrt{3 + \sqrt{5}}}{4\sqrt{5}} > 0.$$

Thus $K(x) \neq 0$ internally, without invoking a ghost identification, and its rank is exactly one. $\square$

Lemma 5 (The TCC coefficient identity). *For the canonical tuple $t = (4, 1, \langle 1, -3, 1 \rangle)$, $\Pi_I^2 = \Pi_I$ is exactly the $\lambda = 1$ instance of AFK equation (1.49), not merely a rank analogy.*

Proof. Expand Π_I^2 using $D_{\mathbf{p}} D_{\mathbf{q}} = \tau^{\langle \mathbf{p}, \mathbf{q} \rangle} D_{\mathbf{p}+\mathbf{q}}$ and compare the coefficient of each $D_{\mathbf{p}}$. Substitute $a_{\mathbf{q}} = \nu_{\mathbf{q}}/\sqrt{5}$ and $\nu_{\mathbf{q}} = \Phi_{\mathbf{q}} \operatorname{shin}_B^{\mathbf{q}/4}(\beta)$. AFK's phase quotient converts the Weyl chirp into

$$\sum_{\mathbf{q} \in \mathcal{I}_{\mathbf{p}}} \tau^{\langle \mathbf{p}, (I+F_t)\mathbf{q} \rangle} \operatorname{shin}_{A_t^{-1}}^{\mathbf{q}/4}(\beta) \operatorname{shin}_{A_t^{-1}}^{(\mathbf{q}-\mathbf{p})/4}(\beta) = 16\delta_{\mathbf{p}, \mathbf{0}}^{(4)},$$

where F_t is AFK's Zauner generator. This is equation (1.49) with $r = 1$ and $\lambda = 1$. Conversely the same coefficient comparison reverses every step, so the equivalence is exact. $\square$

Theorem 6 (Both dimension-four TCC shifts). *Both 0 and 1 are shifts for every dimension-four admissible tuple.*

Proof. Lemma 1 identifies $\Pi_I = K(x)$ and gives $\operatorname{Tr} K(x) = 1$. A rank-one matrix satisfies $K^2 = (\operatorname{Tr} K)K$, so Theorem 4 gives $K(x)^2 = K(x)$. Lemma 5 therefore proves that 1 is a shift for the canonical tuple. The twist-shift congruence is also explicit: for $G = I$ it is $2\lambda + 7 \equiv 1 \pmod{8}$, hence $\lambda \equiv 1 \pmod{4}$.

Complex conjugation sends the twist G to JG , where $J = \operatorname{diag}(1, -1)$, preserves idempotence, and sends a shift to

$$\bar{\lambda} = -(\lambda + d_j - 1) \pmod{d}.$$

Here $d_j = d = 4$, so $\bar{\lambda} = 1 - \lambda \pmod{4}$; consequently the conjugate of the verified $\lambda = 1$ equation is precisely the $\lambda = 0$ equation. Equivalently $\det(JG) = -1$ in the twist-shift congruence gives $-(2\lambda + 7) \equiv 1 \pmod{8}$, hence $\lambda \equiv 0 \pmod{4}$.

It remains to pass from the canonical form to every admissible form. For $d = 4$, admissibility forces $r = 1$, $K = \mathbb{Q}(\sqrt{5})$, and $j = m = 1$. Since $f_1 = 1$, every admissible form has conductor one and discriminant five. Such primitive indefinite forms constitute one proper equivalence class: $h_K = 1$ and the norm- -1 unit ϕ imply that the narrow class number is also one. Thus every form is Q_M for some $M \in \operatorname{GL}_2(\mathbb{Z})$. The transformation formulas of AFK §7.1 change variables $\mathbf{q} \mapsto \ell M \mathbf{q}$ in equation (1.49), where $\ell = \operatorname{sgn} j_{M^{-1}}(\beta)$, and give $\mathcal{Z}_t \subseteq \mathcal{Z}_{t_M}$. Applying the identical argument to M^{-1} gives the reverse inclusion. Therefore both verified shifts belong to every dimension-four admissible tuple. $\square$

7 Scope and review points

Theorems 4 and 6 prove the complete dimension-four instance of the formal Twisted Convolution Conjecture. They do not prove the Minimalist RM Values Conjecture, the Stark conjecture, general TCC, or Zauner’s conjecture in arbitrary dimension.

References

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